

Hypoxia

Hopefully by now we are starting to develop a healthy appreciation for oxygen delivery – why it is so important, how it can be augmented, and the limitations that exist. We have also discussed that when this process starts to break down (i.e., when oxygen delivery is compromised in cardiac/respiratory failure), we must optimize oxygen delivery through critical care interventions, which can come with a cost.

Let's briefly return to our consideration of oxygen delivery and hemoglobin binding. In [Chapter 1](#), we explored how oxygen would bind to hemoglobin and how important the oxygenated hemoglobin molecule was to the delivery of oxygen. Now we will explore what happens when there is less oxygen readily available to be able to bind and saturate hemoglobin ([Fig. 4.1](#)).

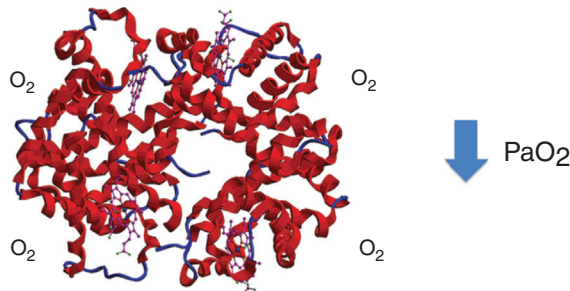


FIG. 4.1 Saturation of hemoglobin requires adequate PaO_2 . (Modified from [Raimundo79/Shutterstock.com](#).)

HOW DOES OXYGEN ENTER THE BODY?

Take a deep breath in. Go ahead, I'll wait ... now let it out. What is happening when we do this simple function? With every breath in and every breath out, we are allowing oxygen to enter our body, we are enabling the delivery and use of that oxygen, and we are expiring carbon dioxide, the byproduct of that metabolism. This happens like clockwork, hundreds of times per hour, every day, whether we are conscious of it or not. None of this would be possible if there was not a ready and efficient way for oxygen to enter the body and carbon dioxide to exit.

To explore how this happens, follow that breath of air in – through the airway, past the bronchi, through the bronchioles, all the way down until we get to the level of the alveoli. It is at this level that gas exchange between the lungs and the blood is able to happen in an efficient manner, by setting up an interface between the blood vessels and the alveoli.

Due to the close proximity of the blood and the alveoli, this interface allows for diffusion of gases down their respective concentration gradients. Just like if you dropped a drop of dye into a glass of water, diffusion allows molecules to go from areas of high concentration to areas of low concentration. It is this diffusion gradient that drives the flow of oxygen and carbon dioxide in such a manner that oxygen is able to enter the body and carbon dioxide is able to leave the body (Fig. 4.2).

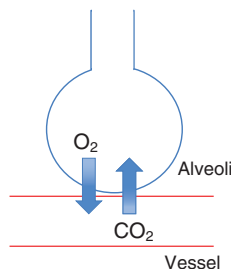


FIG. 4.2 O₂ and CO₂ diffusion at the blood-alveolar interface.

It is important to separate these two components of respiration out (ventilation of the alveoli and perfusion of the blood vessels) because they become vital to our discussion and allow us to better identify the level of the deficiency.

WHEN THE SYSTEM BREAKS DOWN

There are a host of reasons for low concentrations of oxygen in the blood. These can include hypoventilation, low inspired FiO₂, as is seen at high altitude, and diffusion defects where there is a breakdown at the level of the interstitium. The purpose of our discussion is to better understand the two components of the alveolar-blood interface, and how they interact with each other.

Deficiency at the level of the alveolus is labeled V by convention, whereas deficiency at the level of blood to perfuse the alveolus is labeled Q. If both are equally contributing to gas exchange, then the ratio of V/Q is equal to 1 (Fig. 4.3).

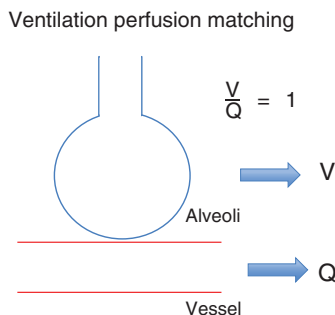


FIG. 4.3 Ventilation perfusion matching.

When there is a deficiency of one relative to the other, there is V/Q mismatch. Let's look at two permutations.

In [Chapter 2](#), we described in detail the specifics of when and how the right heart can fail. Now let's consider the effect of a decline in right heart function on respiratory function.

If there is a decline in the right ventricular function, the overall perfusion to the alveolar-blood vessel apparatus will decline. That means that on some level, whether just through decreased perfusion or the total lack of perfusion, there will be $V/Q > 1$. This manifests as an alveolar unit where is fully ventilated but does not participate in gas exchange. It is as if there is no connection between the two – air comes in and out with no improvement in ventilation. This is referred to as “dead space” (Fig. 4.4).

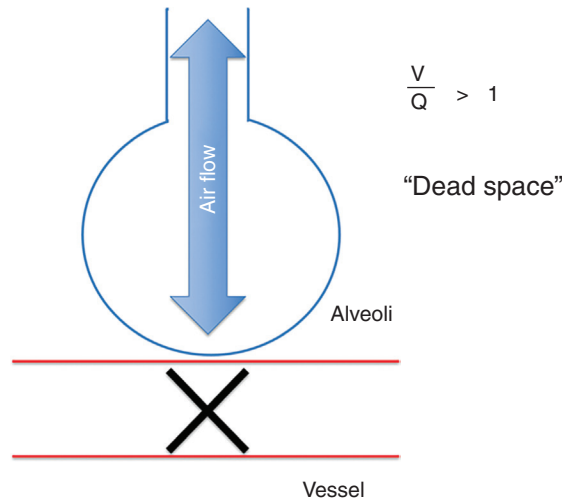


FIG. 4.4 Schematic for dead space

Let's contrast this with the opposite scenario, where $V/Q < 1$. This would correspond with any time when there is decreased ability of air to enter into the alveolar component of the alveolar-blood vessel interface – pneumonia, edema, mucous plugging, etc. In this case, blood traverses across the interface with no ability for gas to diffuse. In this manner, this is referred to as “shunt” (Fig. 4.5).

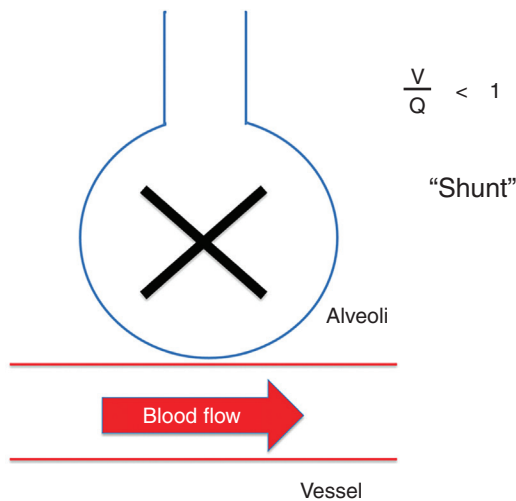


FIG. 4.5 Schematic for shunt

Although the temptation is to think of the lungs as a homogenous collection of alveoli and the pulmonary circulation similarly as a uniform collection of blood vessels perfusing these alveoli, the situation can be quite complex. There can be areas of the lung with profound V/Q mismatch, such as complete collapse of one lobe of the lung due to mucous plug, for example. In this case, the collapsed area of the lung may have a V/Q much less than 1, while the remainder of the lung may have normal matching of ventilation to perfusion.

Consider also that even under normal conditions, blood tends to be gravity dependent leading to increased distension of the pulmonary blood vessels in the lower portions of the lung (and thus $V/Q < 1$), while alveoli may be more distended in the upper areas of the lungs (with $V/Q > 1$) (Fig. 4.6).

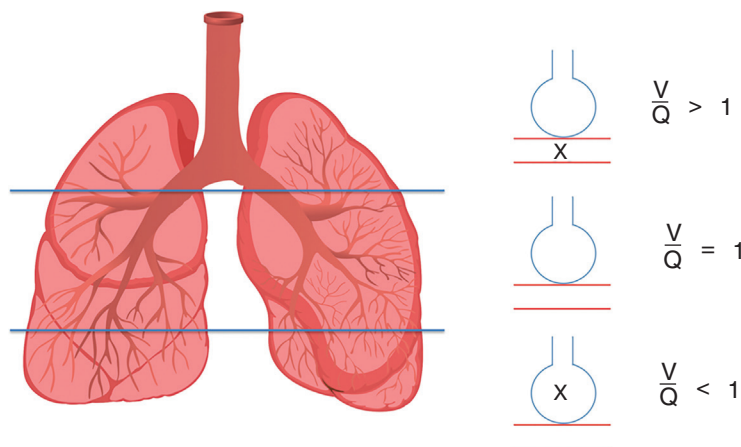


FIG 4.6 Variation in ventilation and perfusion matching in normal lungs. (Modified from YanGe_Tam/Shutterstock.com.)

So why make this distinction?

We distinguish aberrations in the V/Q matching in this way, because the physiology is very different. Just like the models for shock, this classification should help to develop an approach for thinking about respiratory failure, which can then be built upon. Specifically, let's examine the case of hypoxia and the role of mechanical ventilation.

There are many reasons to undergo mechanical ventilation – inability to protect airway, excessive secretions, expected clinical course, etc. However, in order to best understand the breakdown of the delivery of oxygen due to respiratory failure, we will focus our discussion on the role of mechanical ventilation on improving hypoxia due to shunt, where V/Q is less than 1. Let's now focus in on shunt, its specific physiology, and the role of mechanical ventilation in mitigating the ensuing hypoxia.

WHAT DOES HYPOXIA DUE TO SHUNT LOOK LIKE?

Shunt is important to recognize and understand because its physiology is unique.

While we have talked about the alveolar-blood vessel interface as one unit, there are millions of alveoli in the lungs, so when we talk about shunt physiology, we are acknowledging that this physiology exists as a spectrum, becoming more profound as more shunting occurs.

Let's start by understanding what happens with normal lungs, with minimal shunt. If no shunt exists, every alveoli is paired with a perfusing blood vessel and the maximum interface between those two is maintained. If you were to spread out the surface area of this interface, it would be the size of a tennis court, so you can quickly understand how the lungs can be so efficient at allowing oxygen to diffuse into the blood.

Not only are our hypothetical lungs with no shunt able to allow for a higher diffusion of oxygen at room air, but when we add higher concentration of inspired oxygen (FiO_2), we really start to see the difference. Because there is so much more area for oxygen to diffuse across, increasing the FiO_2 by a multiple of two-, three-, or fourfold not only increases the amount of oxygen that diffuses across but it does so at an exponential rate.

Now, let's consider what happens when there is 10% shunt. In this case, there is a slightly lower amount of oxygen that can diffuse across at room air, as noted by the lower PaO_2 . However, as we increase the FiO_2 , say from 21 L nasal cannula to 6 L to a non-rebreather face mask, there is going to be an increase in PaO_2 of the blood almost as high if there was no shunt – however, notice that it takes a higher FiO_2 to get there!

As the amount of shunt progresses, notice that it not only corresponds to a lower initial PaO_2 at room air but also to a more shallow slope of the line, with decreasing gains in PaO_2 for every escalation in FiO_2 (Fig. 4.7).

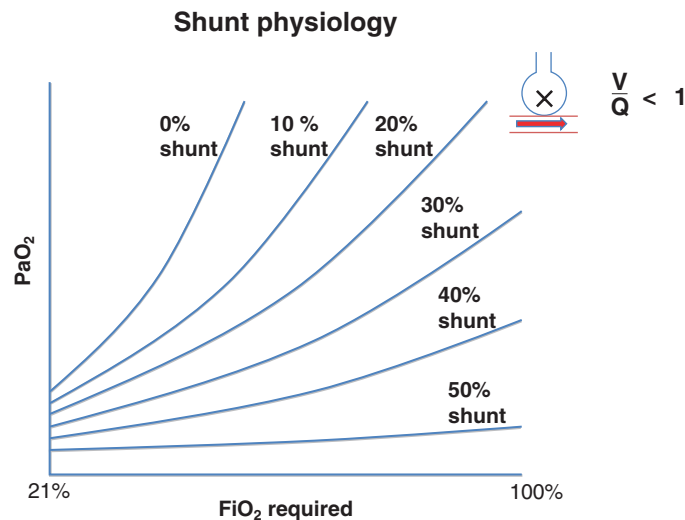


FIG 4.7 Shunt physiology and the relation of PaO_2 to FiO_2

Notice what happens when we arrive at approximately 50% shunt. At this point, you can observe that the curve is essentially flat. This means that no matter how much oxygen you give, there will be no subsequent increase in oxygen that diffuses along to the blood. This pattern is what is meant when we say “shunt physiology” – the point where adding more inspired oxygen will not help increase the blood oxygen concentration, no matter how much is given (Fig. 4.8).

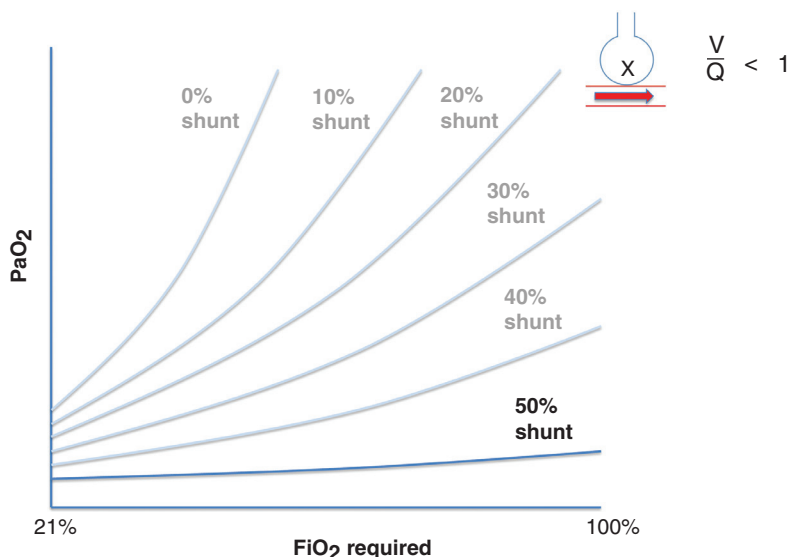


FIG. 4.8 Unresponsiveness to FiO_2 increases with progression of shunt

Why is this?

Imagine you are driving down a road where the speed limit is 30 miles per hour. You don't want to get caught speeding, so you set your speed at 30 miles per hour. You then pass a sign that says the speed limit is now 40 miles per hour. You can now increase your speed to 40 miles per hour.

However, now imagine that you are traveling behind a truck that is going 30 miles an hour. You pass the sign that the speed limit is 40 miles an hour; however, the truck continues driving at 30 miles an hour. Assuming you cannot pass the truck, your speed has to stay at 30 miles per hour.

The same is true of FiO_2 with shunt. The higher the FiO_2 the more oxygen comes across the alveolar-blood interface. However, when alveoli are shunted, the actual inspired amount of oxygen is inconsequential, just like the speed limit is inconsequential if you are stuck behind another car (Fig. 4.7).

So that's it? Game over?

Not necessarily. It just means that we have to improve the shunt fraction if we want to improve the overall oxygenation.

Let's imagine that you are caring for a patient with pneumonia who has become progressively hypoxic over the last several hours. You have witnessed the PaO_2 (and oxygen saturation) continue to decline despite increasing FiO_2 levels, as you have progressed from nasal cannula to non-rebreather. What do you do next?

As you have intuitively come up with, the next step is consideration for intubation. Let's look at why intubation (and positive pressure) improves oxygenation.

IMPACT OF POSITIVE PRESSURE ON HYPOXIA DUE TO SHUNT

Although we initially started discussing shunt as a mismatch between the alveolar unit (V) and the vascular unit (Q), in reality, the lungs exist as a heterogeneous collection of alveoli, which exist in various phases of collapse and inflation.

Positive pressure further inflates the lungs. It pushes air into the lungs, allowing for alveoli that are not open to inflate. This allows for recruitment of alveoli that are not participating in gas exchange (Fig. 4.9A,B).

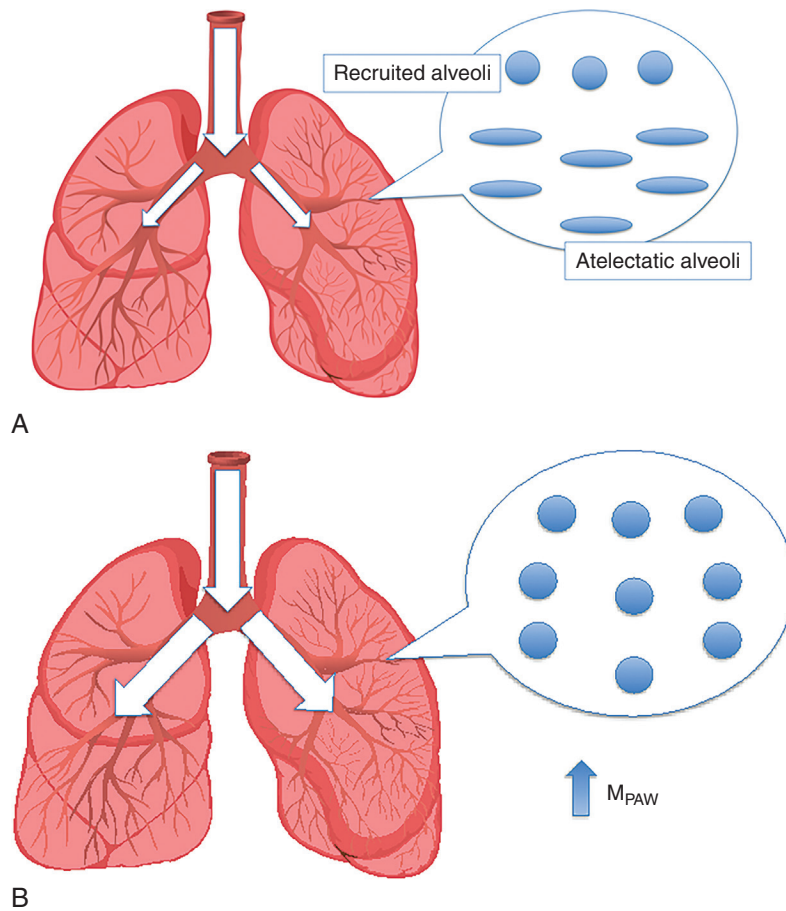


FIG. 4.9 (A) and (B) Positive pressure and alveolar recruitment. M_{PAW} Mean airway pressure. (Modified from [YanGe_Tam/Shutterstock.com](https://www.shutterstock.com/user/YanGe_Tam).)

WHAT IS THE MEAN AIRWAY PRESSURE AND HOW CAN WE AUGMENT IT?

When a breath is delivered from a ventilator, there is a specific amount of pressure that is exhibited at the beginning of a breath and a higher amount of pressure that is exhibited at the end of the breath. The mean pressure is the average pressure exerted between these two pressures and ultimately affects what is exerted on the alveoli. The higher the mean airway pressure, the higher the pressure that can be exerted to open recruitable alveoli (to a point).

The mean airway pressure is ultimately determined by the pressure at the beginning and end of the breath as well as the time spent during inspiration and exhalation, according to the following relationship:

$$M_{PAW} = \frac{(T_i \times P_{IP}) + (T_E \times PEEP)}{T_i + T_E}$$

Where M_{PAW} is the mean airway pressure, P_{IP} is the inspiratory pressure, PEEP is the positive end expiratory pressure, T_i is the time of inspiration, and T_E is the time of expiration.

Since the time of exhalation is greater than the time of inhalation, augmentation of PEEP is the most ready way to augment the mean airway pressure and, by extension, improve shunt and oxygenation.

WHAT ARE THE EFFECTS OF INCREASING PEEP?

Let's envision that shunt continues to worsen on mechanical ventilation, leading us to increase PEEP in order to augment mean airway pressure and recruit additional alveoli into gas exchange. What is the effect of this?

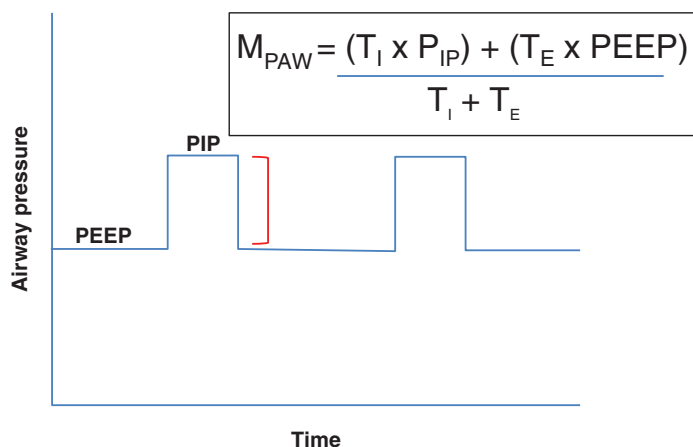


FIG. 4.10 Airway pressure over time

Let's look at the graph in [Fig. 4.10](#), comparing airway pressure over time for a patient on positive pressure ventilation.

You will notice that all airway pressures are positive, with airway pressure going from a low point during the end of expiration (PEEP) to a high point during inspiration (PIP). The difference between these two pressures is the amount of pressure that is delivered with each breath, or the driving pressure.

Now let's explore increasing PEEP. Increasing PEEP will raise the baseline and increase the overall mean airway pressure. However, the effect of doing so may be that either the difference between the two pressures decreases or that the inspiratory pressure must increase if you want to maintain the same driving pressure.

WHAT ARE THE EFFECTS OF WORSENING COMPLIANCE?

This effect on the overall inspiratory pressures becomes more pronounced as compliance worsens. What causes compliance to worsen? Take your pick – inflammation/acute respiratory distress syndrome (ARDS), edema, mucous plugging, atelectasis, pneumonia, etc. Either way, as the alveolar unit becomes progressively flooded and collapsed, two adverse effects occur: the shunt worsens and the lungs become less likely to expand with any breath. Imagine trying to blow up a balloon right out of the package versus one that has been stretched out. The relationship between pressure delivered and compliance is as follows:

$$\text{Compliance} = \frac{\Delta \text{Volume}}{\Delta \text{Pressure}}$$

Which in the context of mechanical ventilation can be rewritten as:

$$\text{Compliance} = \frac{V_T}{\text{PIP} - \text{PEEP}}$$

Where V_T is the tidal volume, or the total volume delivered with each breath.

As compliance decreases, this means that only two things can happen with regards to this relationship – either higher driving pressures have to be used to maintain the same tidal volume, or you will observe lower tidal volume at the same driving pressure.

Maintaining an adequate tidal volume becomes essential, as it is one of the two variables determining the clearance of CO_2 , with the following relationship:

$$\text{Minute Ventilation} = V_T \times \text{Respiratory Rate}$$

THE CHALLENGING TRADEOFF OF WORSENING RESPIRATORY FAILURE

Now you are faced with a difficult tradeoff that often comes up as lungs become progressively compromised. You have to maintain adequate mean airway pressures to allow for oxygenation, but this may lead to either higher inspiratory airway pressures or decreased tidal volume. Higher inspiratory airway pressures will eventually damage the lungs (called barotrauma) and worsen mortality, while lower tidal volumes will lead to a lower minute ventilation and worsening CO_2 clearance. This can be initially mitigated by increasing the respiratory rate to a point but, eventually, you are faced with the situation where you are faced with worsening oxygenation/ventilation or potentially damaging the lungs with higher settings.

PUTTING IT TOGETHER

Maintaining adequate levels of diffused oxygen in the blood is essential for hemoglobin to adequately saturate. Without well-saturated hemoglobin, oxygen delivery becomes a progressively tenuous proposition.

While there are many causes of hypoxia, shunt is an important cause, as there is a limited effect of increasing inspired oxygen (FiO_2). Rather, alveolar units must be recruited through increasing the mean airway pressure. However, as respiratory failure continues to worsen, with progression of shunt fraction and worsening compliance, the choice sometimes becomes either accepting worsening parameters of oxygenation/ventilation or subjecting the lungs to more damaging effects as we continue to increase the “dose” of mechanical ventilation. We will further explore the effects of how we “dose” mechanical ventilation and other critical care interventions in the next chapter.

SUGGESTED READING

- Benatar, S. R., Hewlett, A. M., & Nunn, J. F. (1973). The use of iso-shunt lines for control of oxygen therapy. *British Journal of Anaesthesia*, 45(7), 711–718.
- Doorduyn, J., Nollet, J. L., Vugts, M. P., Roesthuis, L. H., Akankan, F., & van der Hoeven, J. G., et al. (2016). Assessment of dead-space ventilation in patients with acute respiratory distress syndrome: a prospective observational study. *Critical Care*, 20(1), 1–10.
- Henderson, W. R., & Sheel, A. W. (2012). Pulmonary mechanics during mechanical ventilation. *Respiratory Physiology & Neurobiology*, 180(2-3), 162–172.

- Hess, D. R. (2015). Recruitment maneuvers and PEEP titration. *Respiratory Care*, 60(11), 1688–1704.
- Malhotra, A. (2007). Low-tidal-volume ventilation in the acute respiratory distress syndrome. *New England Journal of Medicine*, 357(11), 1113–1120.
- Marini, J. J., & Ravenscraft, S. A. (1992). Mean airway pressure: physiologic determinants and clinical importance—Part 1: physiologic determinants and measurements. *Critical Care Medicine*, 20(10), 1461–1472.
- Mercat, A., Richard, J. C., Vielle, B., Jaber, S., Osman, D., Diehl, J. L., et al. (2008). Positive end-expiratory pressure setting in adults with acute lung injury and acute respiratory distress syndrome: a randomized controlled trial. *JAMA*, 299(6), 646–655.
- West, J. B. (2012). *Respiratory physiology: The Essentials*. Lippincott Williams & Wilkins.